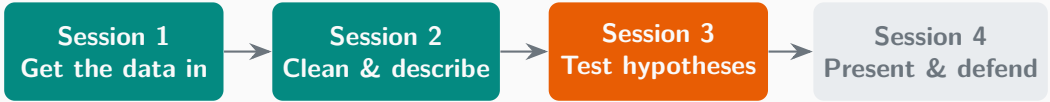
Python for Data Science

Session 3 — Testing Hypotheses

Andrea Guido

PSB Paris School of Business — PGE 5

Last time the data showed three patterns; today we test them



Where we left off: a cleaned dataset, a cleaning log, and three untested patterns: default rates differ by education; defaulters have lower limits; late months go with default.

Today's question for each pattern: **could it be chance, how big is it, and what does it mean?**

Pick 100 clients at random and compare the credit limits of defaulters and non-defaulters.

57%

of the time, the test says the gap could be chance.

With all 30,000 clients: **never**.

Same data, same gap, different verdict. Today you learn why.

Today: one test per research question, then the limits of what tests can say

1. **From question to hypothesis**: what exactly are we testing?
2. **Comparing two groups** (RQ2): t-test, a non-parametric alternative, effect size
3. **Categorical data** (RQ1): chi-square, and the black lines on your Session 2 bars
4. **Linear regression** (RQ3): many variables at once
5. **Critical interpretation**: significance vs importance, correlation vs cause, AI tools

Before class you watched the 20-minute refresher on hypothesis testing and regression. We apply it to our data.

Part 1: From question to hypothesis

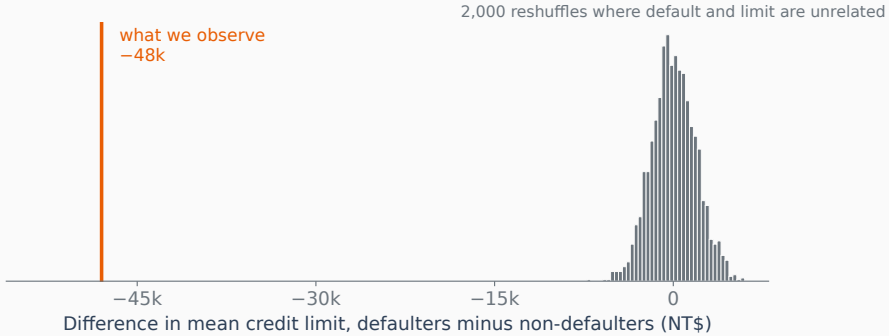
A test needs a precise claim to reject

Each research question becomes a null hypothesis we try to reject

	Null hypothesis H_0 ("nothing going on")	Test
RQ1	the default rate is the same at every education level	chi-square
RQ2	defaulters and non-defaulters have the same mean limit	t-test
RQ3	late months do not predict default once limit, age, sex and education are held fixed	regression

The type of variables decides the test: categories vs categories → chi-square; a number across two groups → t-test; many variables → regression.

A test asks: if nothing were going on, how surprising would our data be?



Shuffle the default labels at random 2,000 times: the gap in limits never goes beyond ±7k. The real gap is **−48k**. Chance cannot produce it.

The p-value is the share of "nothing going on" worlds at least as extreme as ours

What it is

The probability of a result at least this extreme **if H_0 were true**. Small p: our data would be very surprising under H_0, so we reject it. Convention: $p < 0.05$.

What it is not

Not the probability that H_0 is true. Not the size of the effect. Not a measure of importance. A p-value of 0.001 can go with a tiny, useless difference.

Why it matters: most misreadings of statistics in finance and in the press confuse "significant" with "large".

Part 2: Comparing two groups (RQ2)

Do defaulters have lower credit limits?

Welch's t-test: the gap is 48k, and it is not chance

```
from scipy import stats

yes = df.loc[df["default"] == 1, "limit"]
no  = df.loc[df["default"] == 0, "limit"]
res = stats.ttest_ind(yes, no, equal_var=False)   # Welch
res.statistic, res.pvalue, res.confidence_interval()
```

```
(-28.95, 3.4e-178, ConfidenceInterval(low=-51239, high=-44741))
```

`equal_var=False`: do not assume both groups have the same spread (they do not). The **confidence interval** is the useful part: the true gap is most likely between −51k and −45k.

The effect size puts the gap on a scale you can compare

```
pooled_sd = np.sqrt((yes.var() + no.var()) / 2)
d = (yes.mean() - no.mean()) / pooled_sd
```

```
-0.39
```

Cohen's d (rule of thumb)	0.2	0.5	0.8
	small	medium	large

The gap is about a third of a standard deviation: **between small and medium**. Real, but the two distributions overlap a lot (recall the Session 2 boxplot).

Limits are skewed, so check with a test that does not assume normality

```
res = stats.mannwhitneyu(yes, no)
res.pvalue
```

```
1.2e-189
```

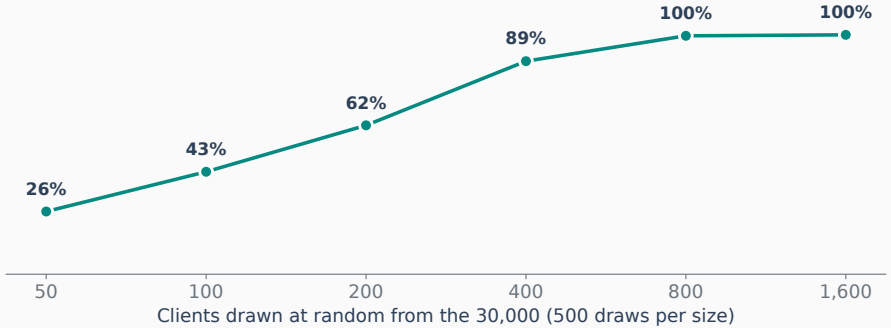
Mann–Whitney

Compares **ranks**, not means. Asks: does a random defaulter tend to have a lower limit than a random non-defaulter?

Why run both

If both tests agree, the conclusion does not depend on the normality assumption. Here they agree.

With 30,000 clients, almost any difference becomes significant



Share of random samples in which the limit gap is significant ($p < 0.05$). The gap is the same in every sample; only the sample size changes. **Significance measures precision, not importance.**

Part 3: Categorical data (RQ1)

Does the default rate differ across education levels?

The chi-square test compares observed counts with “no relationship” counts

```
table = pd.crosstab(df["education"], df["default"])
res = stats.chi2_contingency(table)
res.statistic, res.pvalue, res.dof
```

```
(116.9, 3.7e-25, 3)
```

Defaulters	Observed	Expected if H_0 true
Graduate school	2,036	2,359
University	3,330	3,127
High school	1,237	1,096
Others	7	27

Significant, but weak: Cramér's V is 0.06

```
n = table.values.sum()
v = np.sqrt(res.statistic / (n * (min(table.shape) - 1)))
```

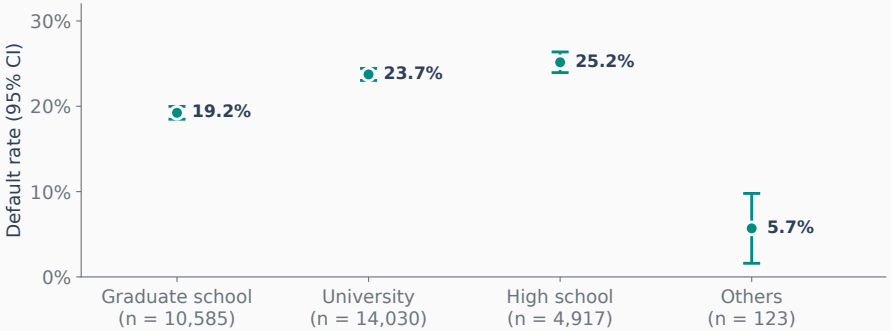
```
0.063
```

Cramér's V runs from 0 (no association) to 1 (perfect). Below 0.1 is usually called **negligible**.

The p-value says the rates are not all equal. V says education tells you **very little** about who defaults.

Why it matters: "education significantly affects default" and "education barely matters" are both true here. Which sentence you write is a choice.

The black lines on your Session 2 bars were 95% confidence intervals

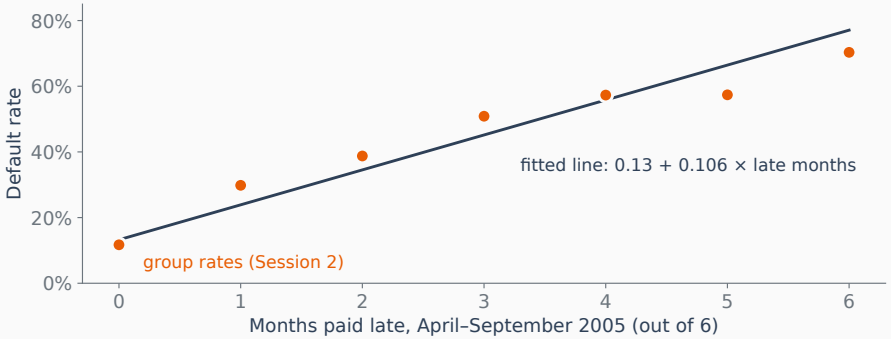


With 10,000+ clients the interval is tiny. “Others” rests on 123 clients: its interval spans 2% to 10%. The smaller the group, the less you know.

Part 4: Linear regression (RQ3)

Many variables at once

A regression line summarises the Session 2 chart in two numbers



Intercept 0.13: the default rate with no late months. Slope 0.106: each extra late month adds about **10.6 percentage points**.

One line of `statsmodels` fits the model; the formula reads like the question

```python
import statsmodels.formula.api as smf

df["limit10k"] = df["limit"] / 10_000
model = smf.ols("default ~ late_months + limit10k + age"
                " + C(sex) + C(education)", data=df)
res = model.fit(cov_type="HC1")      # robust standard errors
res.summary()
```

- `default ~ ...`: outcome on the left, explanatory variables on the right
- `C(...)`: treat as categories. One level becomes the reference group
- `limit10k`: rescaled so the coefficient reads "per NT$10,000"

Read the output one row at a time: each coefficient holds the others fixed

Variable	Coefficient	95% CI	Reading
Late months (per month)	0.102	[0.099, 0.105]	+10 pp per late month
Credit limit (per 10k)	−0.002	[−0.002, −0.002]	−2 pp per 100k
Age (per year)	0.001	[0.001, 0.002]	+1 pp per decade
Male (vs female)	0.016	[0.007, 0.025]	+1.6 pp
University (vs graduate)	0.005	[−0.005, 0.014]	no clear difference
High school (vs graduate)	0.000	[−0.014, 0.014]	no clear difference
Others (vs graduate)	−0.083	[−0.125, −0.041]	123 clients only

Linear probability model, robust standard errors. $N = 29{,}655$ (345 unknown education codes excluded). $R^2 = 0.16$.

RQ1 revisited: once repayment is known, education adds almost nothing

Without controls (chi-square)

High-school clients default 6 pp more often than graduates.

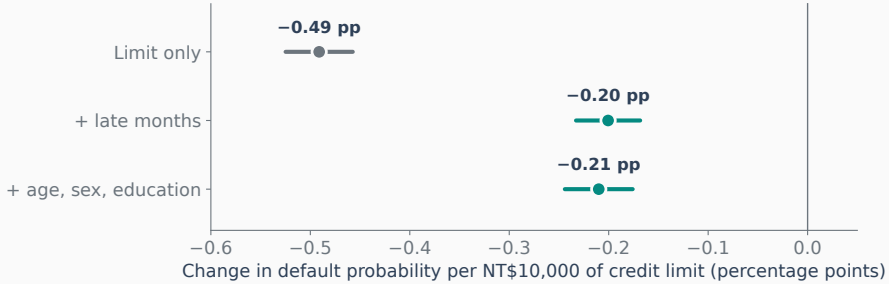
With controls (regression)

Same repayment record, limit, age and sex: the gap is 0.0 pp.

Education goes with repayment behaviour and credit limits; those, not education itself, carry the information. The only "significant" category, *others*, rests on 123 clients whose schooling we cannot even name.

Why it matters: a control variable can make a pattern disappear. Always ask what else differs between the groups you compare.

RQ2 revisited: the limit effect shrinks by 60% once repayment history enters



The bank sets lower limits for clients it sees as risky. Part of the raw gap was the bank's own risk assessment, as the Session 1 diagram warned.

$R^2 = 0.16$: we explain a sixth of the variation in default

What the model gets

Repayment history is by far the strongest signal. Limit alone gives $R^2 = 0.02$; adding late months raises it to 0.16.

What it misses

Income, job loss, other debts: none are in the data. Most of what decides default is invisible to us.

Why it matters: "the bank could have seen it coming" is only partly true. It could see who was *more likely* to default, not who *would*.

Check the residuals: a linear model can predict impossible probabilities

```
pred = res.fittedvalues
(pred < 0).sum(), pred.min().round(3), pred.max().round(3)
```

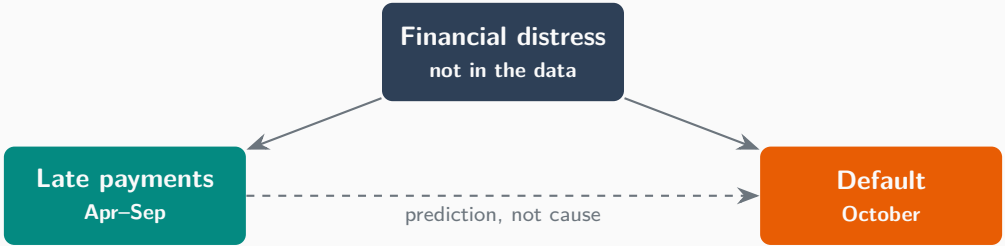
```
(8, -0.073, 0.826)
```

- Eight clients get a predicted default probability below zero
- Residuals (`res.resid`) of a 0/1 outcome only take two values per prediction: plot them against `pred` to see it
- Coefficients near the middle of the data are still reliable; for predictions, a **logit** model (`smf.logit`) keeps them between 0 and 1

Part 5: Critical interpretation

What your results can and cannot say

Late payments predict default, but they do not cause it



A client who lost their job pays late *and* then defaults. Stopping them from paying late would not prevent the default. For a bank's **prediction** task that does not matter; for a **policy** question it matters a lot.

The data describe one bank, in one country, in one unusual year

One bank

Its own clients and its own rules for setting limits.

Taiwan, 2005

In the middle of a credit card debt crisis: default was unusually common.

Six months

One outcome month. We know nothing about what happened next.

Why it matters: a result can be precise and still not generalise. Say where it applies.

Use AI tools, but you answer for every line they write

Verify

Run the code and read it line by line. A common slip: `ttest_ind` without `equal_var=False`, or `PAY_0` taken as April.

Sanity-check

Compare the output with a simple `groupby`. If a regression says the opposite of the raw means, find out why before you report it.

Disclose

State in your notebook which tool you used and for what. At the oral exam, "the AI wrote it" is not an answer.

Our three answers, each with its evidence and its limit

	Answer	Evidence	Limit
RQ1	rates differ, but education adds almost nothing once repayment is known	χ^2: $p < 0.001$, $V = 0.06$; regression: ≈ 0	"others" is only 123 clients
RQ2	yes, 48k lower on average	Welch and Mann–Whitney, $d = -0.39$	reflects the bank's choice
RQ3	yes: $+10$ pp per late month, with controls	regression, CI [0.099, 0.105]	prediction, not cause

In-class exercise, and your presentation for Session 4

Today (feeds your presentation)

1. State H_0 for your research question
2. Run the test that fits it, on **your** cleaned data
3. Report the p-value, the effect size and a confidence interval
4. Write three sentences: what it means, how big it is, one limit

For Session 4 (10 minutes per group)

1. The question and your answer, first
2. Your key cleaning decisions and why
3. One plot and one test, interpreted
4. The main limit of your answer

Work in `session3.ipynb`. Presentation slides are due **before Session 4** (50% of the grade). Every member of the group must be able to explain every line.

A p-value tells you whether a pattern could be chance, not whether it matters or why it exists.

Next session: you defend your answer, and see that other groups reached different ones.